FENG-AN (Alan) HSIEH

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Summary

ESL Modeling Engineer with 4 years of experience building C++/SystemC TLM-2.0 functional models (LT / behavior-level) and integrating them into Synopsys VDK virtual platforms for enterprise and client SSD controllers. End-to-end owner of 6 peripheral and controller IP models, delivered to firmware teams and external customers to enable software bring-up ahead of silicon. Python automation builder for trace analysis, regression, and model delivery.

Experience

SiliconMotion – ESL Modeling Team **Zhubei, Hsinchu**
Chief Engineer *July 2026 – Present*

- Architect and maintain 6 C++/SystemC TLM IP models (4 enterprise, 2 client), including the shared inter-engine messaging IP and core NAND Flash Controller (register map, interrupt, arbitration, DMA, command decode).
- Profiled the messaging IP model with VDK top-down profiling and Valgrind, and replaced per-clock-cycle `wait()` polling with coarser time-based waits, reducing scheduler wake-ups and speeding up simulation.
- Deliver verified virtual platform releases and model documentation to 5+ firmware engineers and external customers; mentor engineers on ESL design patterns and TLM-2.0 modeling standards.
- Built agentic developer workflows (Claude Code, Cursor) for model boilerplate generation, spec-to-model scaffolding, and automated regression, reducing new NAND model delivery from $\sim$2 weeks to 2–3 days.

Senior Engineer *November 2022 – June 2026*

- Integrated peripheral TLM models with processor fast models into full-system virtual platforms on Synopsys VDK, enabling firmware teams to bring up software months ahead of silicon tape-out.
- Conducted hardware-software co-simulation with DV test suites; verified CModels against RTL specs to isolate and resolve complex boundary bugs across HW and FW.
- Automated RTL waveform parsing (FSDB/VCD) with Python and shell scripts, speeding up test creation by 3x and saving $\sim$1 week of manual effort per cycle.
- Built internal Python utilities for platform trace analysis and debugging, accelerating issue isolation for firmware teams.

Skills

Languages & Modeling: C/C++ (C++14), SystemC, TLM-2.0 (LT / behavior-level modeling), Python, Shell, Verilog, Assembly
EDA & Virtual Platforms: Synopsys VDK, Virtual Prototyping, HW/SW Co-simulation, Linux, Git
Protocols & Architecture: AMBA spec & bus integration (AXI, AHB, APB, CHI), NAND Flash Controller Architecture, DMA, Register Maps, Interrupts, State Machines
Engineering Tooling: Debugging (gdb, Platform Trace Analysis), Automated Regression, Claude Code, Cursor, Waveform Parsing (FSDB/VCD)
Languages: Mandarin (Native), English (TOEIC 780)

Education

National Chung Cheng University **Chiayi, Taiwan**
M.E. in Electrical Engineering *August 2020 – July 2022*
Thesis: An End-To-End System For Road Agent Behavior Classification (CNN, LSTM, Object Detection, Tracking)
National University of Kaohsiung **Kaohsiung, Taiwan**
B.E. in Computer Science and Information Engineering *September 2016 – June 2020*